LEAKLISTEN · REGULATORY REGISTER

LeakListen regulatory and standards register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-01	Amish Chadha	First register, written in the kit 1.9.1 upgrade for option A (the current design) and, where it differs, options B and C; the first candidate region is the UK (LKL-DEC-001, 2026-10-02)

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LeakListen as a prototype is a battery-powered licence-exempt radio device that sits in a water utility's valve chamber, so the rules that matter are radio (G1, G2), lithium cell safety and transport (G3), safe working at the chamber (G5, G6) and, for option B only, products in contact with drinking water (G4). No certification is needed before TRL 4; a deployed product would need radio conformity (G2). The first candidate region is the UK (LKL-DEC-001, 2026-10-02), so UK rules lead. Not legal advice; anything marked "to assess" needs a qualified reviewer. Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	UK Interface Requirement 2030, licence-exempt short range devices (Ofcom), and the Wireless Telegraphy exemption regulations	UK	Frequency, power and duty cycle of licence-exempt devices in the 868 MHz band	The LoRaWAN radio at +14 dBm ERP; PR-020	Design lead	Applies (limits to confirm against the current text)
G2	EU Radio Equipment Directive 2014/53/EU with ETSI EN 300 220 (short range devices, 25 to 1000 MHz) and the EMC standards for radio equipment; UK Radio Equipment Regulations 2017	EU, UK	Radio conformity, CE or UKCA marking, spectrum use	A product with the STM32WL radio module; prototype use at TRL 3 needs no marking	Design lead	To assess (certification needed later)
G3	UN3090 and UN3091 (lithium metal batteries, alone and packed with equipment) and UN Manual of Tests and Criteria section 38.3; IEC 60086-4 (safety of lithium primary batteries)	International	Transport classification and cell safety	The Li-SOCl2 C cell (BOM line 8) ships as a dangerous good; fuse and reverse protection (PR-017)	Design lead	Applies (to assess for shipping)
G4	Regulation 31, Water Supply (Water Quality) Regulations (England and Wales), approval of products in contact with drinking water (DWI, WRAS)	UK	Materials and products touching drinking water	Option B only: stainless cap, EPDM gasket, potted hydrophone and cable are wetted. Option A touches no water (PR-019); option C is buried	Co-design partner	To assess (option B only)
G5	Confined Spaces Regulations 1997 (SI 1997/1713) and HSE Approved Code of Practice L101	UK	Work in confined spaces, permits, gas testing, rescue	The design avoids entry (PR-016); the crew's practice at the chamber remains the utility's	Co-design partner	Applies (design intent: no entry)
G6	Safety at Street Works and Road Works, a code of practice (UK)	UK	Signing, lighting and guarding of work in the street	Lifting a cover and working at a chamber in a street	Co-design partner	To assess
G7	IEC 60529, degrees of protection provided by enclosures (IP code)	International	IP68 definition	The logger housing (PR-011); the test is TRL 4 work	Design lead	Applies
G8	UK GDPR and Data Protection Act 2018 (and EU GDPR)	UK, EU	Processing of personal data	A vibration sensor in the street is a poor microphone and raw samples never leave	Design lead	To assess

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
				the device (PR-008), so no personal data is expected		
G9	EU Batteries Regulation (EU) 2023/1542, WEEE Directive 2012/19/EU, RoHS Directive 2011/65/EU	EU	Battery labelling and end of life, electronics waste, hazardous substances	The cell and the board of a product sold in the EU; label (PR-018)	Design lead	To assess (certification needed later)
G10	IATA Dangerous Goods Regulations, magnetised material (Packing Instruction 953)	International	Air shipment of strong magnets	The 42 mm pot magnet, shipped with its keeper plate	Design lead	To assess
G11	ISO 11228-1, manual handling: lifting and carrying	International	Mass limits for lifting	Covers (30 kg or more) and, for option B, the 6 kg cap; the logger set is 0.96 kg	Co-design partner	To assess
G12	Consent to place equipment on a utility's assets, a hydrant or a street verge	Country and utility	Permission	Every installation (option C also needs consent to dig and a services search)	Co-design partner	To assess

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- Ofcom, "IR 2030 UK Interface Requirements 2030", https://www.ofcom.org.uk/__data/assets/pdf_file/0028/84970/ir-2030.pdf, listed in search results 2026-11-01; its text was not consulted for this register (G1).
- Ofcom, "Short-range devices", https://www.ofcom.org.uk/spectrum/radio-equipment/short-range-devices, listed in search results 2026-11-01 (G1).
- Drinking Water Inspectorate, "What is Reg31?", https://www.dwi.gov.uk/?p=8284, listed in search results 2026-11-01; not consulted (G4).
- legislation.gov.uk, "The Confined Spaces Regulations 1997", https://www.legislation.gov.uk/uksi/1997/1713/section/1, and HSE, "Safe work in confined spaces" (L101), https://www.hse.gov.uk/pubns/books/l101.htm, listed in search results 2026-11-01; not consulted (G5).
- Standards G2, G3, G6 to G11 are cited by number and title only, from the author's knowledge; their texts were not consulted for this register and each applicability is to be confirmed by the role named. No standard number was invented; any number the author could not confirm is left out.